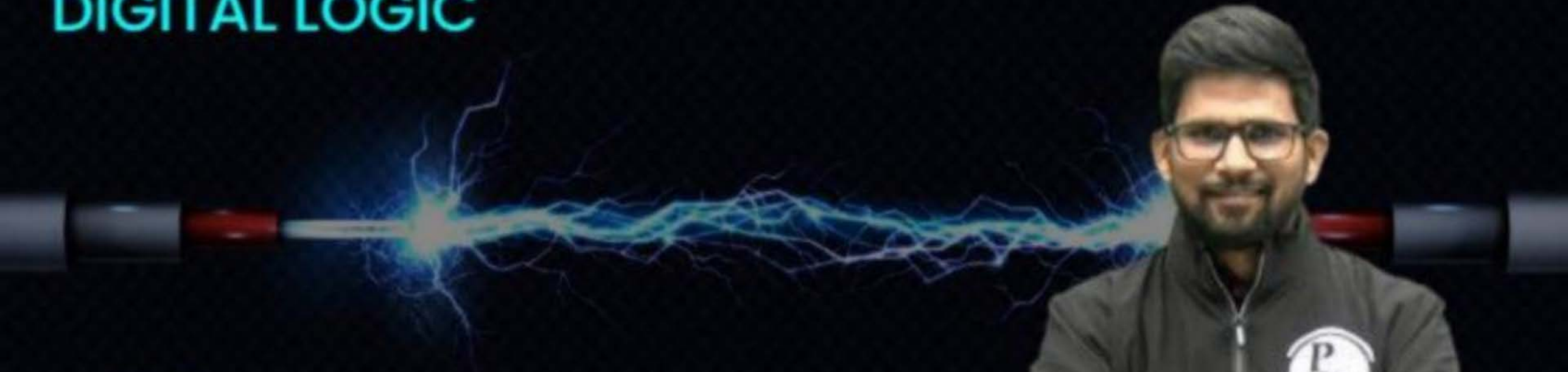
COMPUTER SCIENCE & IT

DIGITAL LOGIC



Lecture No. 06

BOOLEAN THEOREMS AND GATES



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XOR & XNOR GATE



- XOR & XNOR GATE Cont •

- H.W

Q.1 $A \oplus \bar{A} \oplus A \oplus \bar{A} \oplus A \oplus \bar{A} \oplus A = ?$ Ans. $\bar{A}$

Q.2 $(A \oplus B \oplus \bar{C}) \oplus (\bar{A} \oplus B) \oplus (B \oplus \bar{C}) = ?$ Ans. $\bar{B}$

Q.3 $1 \oplus A \oplus 0 \oplus \bar{A} \oplus \bar{B} \oplus 1 \oplus 0 = ?$ Ans. B

Q.4 $(\bar{A} \oplus \bar{B}) \oplus (\bar{B} \oplus \bar{C}) \oplus (\bar{A} \oplus \bar{C}) \oplus (A \oplus B \oplus C) = ?$
Ans. $(A \oplus B \oplus C)$

Q.5 $AB \oplus BC = ? \Rightarrow B[A \oplus C]$

Q.6 $AB \odot BC = ? \Rightarrow AB \odot BC = \bar{B} + A \odot C$

- $AB \oplus BC = \overline{AB} \cdot BC + AB \cdot \overline{BC}$

$= (\overline{A}+\overline{B})BC + AB(\overline{B}+\overline{C})$

$= \overline{A}BC + 0 + 0 + AB\overline{C}$

$= \overline{A}BC + AB\overline{C}$

$= B[\overline{A}C + A\overline{C}]$

$$\boxed{AB \oplus BC = B[A \oplus C]}$$

$$AB \oplus BC = B \cdot [A \oplus C]$$

$$AB \oplus BC \oplus ABC = B \cdot [A \oplus C \oplus AC]$$

$$\overline{A \oplus B} = A \odot B$$

$$\overline{A \odot B} = A \oplus B$$

- $AB \odot BC = \overline{AB} \cdot \overline{BC} + AB \cdot BC$

$= (\overline{A}+B)(\overline{B}+\overline{C}) + ABC$

$= [\overline{B} + \overline{A} \cdot \overline{C}] + ABC$

$= \overline{B} + \overline{A}\,\overline{C} + B \cdot AC$

$= (\overline{B}+B) \cdot (\overline{B}+AC) + \overline{A}\,\overline{C}$

$= \overline{B} + AC + \overline{A}\,\overline{C}$

$= \overline{B} + A \odot C$

$$\boxed{AB \odot BC = \overline{B} + A \odot C} \neq B(A \odot C)$$

$$\overline{AB \oplus BC} = \overline{B \cdot [A \oplus C]} = \overline{B} + A \odot C$$

- $A \oplus B \oplus C = \overline{A \odot B \odot C}$ → true or false

	A (2^2)	B (2^1)	C (2^0)	$y_1 = A \oplus B \oplus C$	$y_2 = A \odot B \odot C$
0	0	0	0	0	0
1	0	0	1	1	1
2	0	1	0	1	1
3	0	1	1	0	0
4	1	0	0	1	1
5	1	0	1	0	0
6	1	1	0	0	0
7	1	1	1	1	1

$y_1[A,B,C] = \Sigma(1,2,4,7)$, $y_2[A,B,C] = \Sigma(1,2,4,7) = A \oplus B \oplus C = A \odot B \odot C$

$1 \oplus 0 \oplus 1 = 0$

$1 \oplus 1 \oplus 1 = 1$

$1 \oplus 0 \oplus 0 = 1$

$1 \oplus 1 \oplus 0 \oplus 0 = 0$

$1 \oplus 0 \oplus 1 \oplus 1 = 1$

$1 \oplus 1 \oplus 1 \oplus 1 = 0$

$1 \oplus 0 \oplus 1 \oplus 0 \oplus 1 = 1$

$1 \odot 0 \odot 0 \odot 1 = 0$

$1 \odot 1 \odot 1 = 1$

$1 \odot 0 \odot 0 = 1$

$1 \odot 1 \odot 0 \odot 0 = 1$

$1 \odot 0 \odot 0 \odot 1 \odot 1 = 0$

$1 \odot 1 \odot 1 \odot 1 = 1$

From above we conclude that :

- XOR gate search for → always odd no. of '1's
  - ↳ if total no. of 1's odd then result is '1'
    otherwise result is '0'.

- XNOR gate search for →

→ odd no. of 1's for odd no. of i/p lines
→ even no. of 1's for even no. of i/p lines

and that's why for odd no. of i/p lines XOR & XNOR are same i.e

$A \oplus B \oplus C = A \odot B \odot C$, $A \oplus B \oplus C \oplus D \oplus E = A \odot B \odot C \odot D \odot E$

while for even no. of i/p lines XOR & XNOR are compliment of each other: i.e.

$$A \oplus B = \overline{A \odot B}$$

$$A \oplus B \oplus C \oplus D = \overline{A \odot B \odot C \odot D}$$

- Compliment property of XOR and XNOR GATE :

$$\overline{A \oplus B} = \overline{A} \oplus B = A \oplus \overline{B} = A \odot B$$

$$\overline{A \odot B} = \overline{A} \odot B = A \odot \overline{B} = A \oplus B$$

$$\overline{A} \oplus \overline{B} = A \odot \overline{B} = A \oplus B$$

$$\overline{A} \odot \overline{B} = A \oplus \overline{B} = A \odot B$$

$$\overline{A \oplus B \oplus C} = \overline{A} \oplus B \oplus C = A \oplus \overline{B} \oplus C = A \oplus B \oplus \overline{C}$$

$$\overline{A \oplus B \oplus C} = A \odot B \oplus C = A \oplus B \odot C$$

$$\overline{\overline{A \oplus B \oplus C}} = \overline{A \odot B \oplus C} = \overline{A} \odot B \oplus C = A \odot \overline{B} \oplus C = A \odot B \oplus \overline{C}$$

$$= A \oplus B \oplus C = A \odot B \odot C$$

$$\overline{A \oplus \overline{B} \odot C \oplus \overline{D}} = \overline{A \oplus B \odot C \oplus D}$$

$$\overline{A \oplus B \odot C \oplus D} = A \oplus B \oplus C \oplus D$$

$$= A \odot B \odot C \oplus D$$

$$= A \oplus B \odot C \odot D$$

$$\overline{A \oplus \overline{B} \oplus \overline{C} \oplus \overline{D}} = \overline{A \oplus B \oplus C \oplus D}$$

$$= A \odot B \oplus C \oplus D = A \odot B \odot C \odot D$$

$$= A \oplus B \odot C \oplus D$$

$$= A \oplus B \oplus C \odot D$$

- $\overline{\overline{A \oplus \overline{B} \oplus C \oplus \overline{D}}} = (A \oplus B \oplus C \oplus D)$
- $\overline{\overline{A \odot B} \oplus \overline{C} \odot \overline{D}} = \overline{A \odot B \oplus C \odot D}$

$= A \odot B \odot C \odot D$

$= A \oplus B \oplus C \odot D$

$= A \odot B \oplus C \oplus D$

$\underline{AB} \oplus \underline{\overline{B} \cdot C} \neq \overline{AB \oplus BC}$

$\underline{AB} \oplus \overline{BC} = \overline{\underline{AB} \oplus BC}$

$\overline{\underline{AB} \oplus (A+B)} = \overline{AB} \oplus (A+B)$

$= AB \oplus \overline{(A+B)}$

$AB \oplus \overline{B}\overline{C} \neq \overline{AB \oplus BC}$

$AB \oplus \overline{(B+C)} = \overline{AB \oplus (B+C)}$

$$A B \oplus \overline{B \cdot C} \neq A B \oplus B C$$

$$A B \oplus \overline{(B+C)}$$

$$= P \oplus \overline{Q}$$

$$= \overline{P \oplus Q}$$

$$= \overline{A B \oplus (B+C)}$$

H.W

1. $\overline{AB \oplus BC} = ?$

2. $A \oplus \overline{B} \oplus \overline{C} \oplus \overline{D} = ?$

3. $AB \oplus \overline{B}C = ?$

4. $AB \oplus \overline{BC} = ?$

5. $\overline{A}B \oplus A\overline{B} = ?$

6. $\overline{A}\,\overline{B} \oplus AB = ?$



2 Minute Summary

XOR & XNOR.

Thank you

GW

Soldiers !

